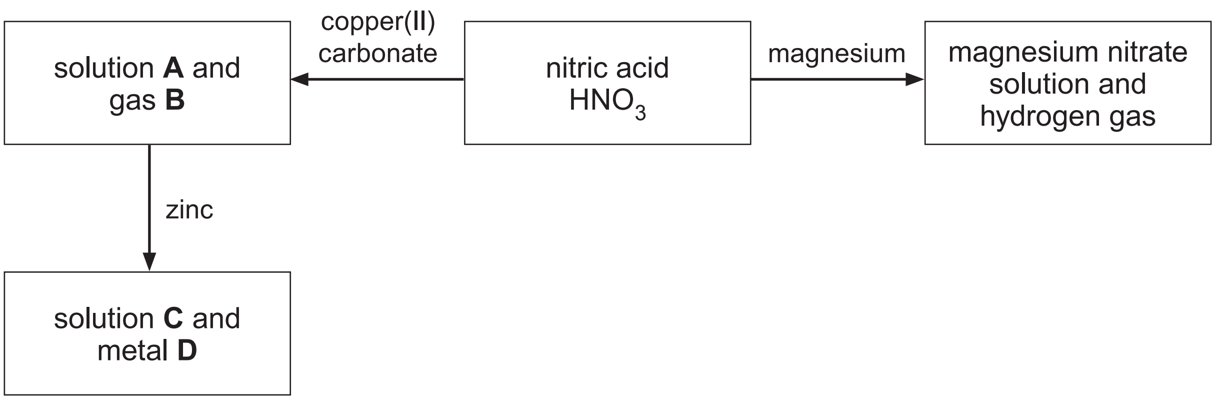
Examiner only

5. (a) The diagram shows a sequence of reactions from nitric acid.



(i) Use the information in the diagram to name the following substances. [3]

solution **A** ....

gas **B** ....

solution **C** ....

metal **D** ....

(ii) Write the balanced symbol equation for the reaction between magnesium and nitric acid. [3]

....

(iii) Name the type of reaction that takes place between solution **A** and zinc. [1]

....

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(b) (i) The sulfate ions in sulfuric acid are identified using barium chloride solution. The equation for the reaction is shown below.

$$H_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(s) + 2HCl(aq)$$

Tick (✓) the correct ionic equation for the formation of the precipitate. [1]

$H^+ + Cl^- \longrightarrow HCl$	☐
$Ba^{2+} + SO_4^{2-} \longrightarrow BaSO_4$	☐
$Ba^{2+} + S^{2-} + 4O^{2-} \longrightarrow BaSO_4$	☐
$2H^+ + 2Cl^- \longrightarrow 2HCl$	☐
$Ba^{2-} + SO_4^{2+} \longrightarrow BaSO_4$	☐

(ii) A student was asked to identify the ions present in barium chloride solution.

I. Other than adding sulfate ions, state how the student would test for barium ions. Give the expected observation for a positive test. [1]

....

....

II. State how the student would test for chloride ions. Give the expected observation for a positive test. [1]

....

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